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QUIZ 8-A (EE-1005)

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12

1. Draw Truth table for JK-FF and generate Excitation table for JK-FF and perform all required steps? [2]

Truth Table:

Q_t	J	K	Q_{t+1}	$\bar{Q}_{t+1}$
0	0	0	0	1
0	0	1	0	1
0	1	0	1	0
0	1	1	1	0
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	0

Excitation table:

Q_t	Q_{t+1}	J	K
0	0	0	0
0	1	1	0
1	0	0	1
1	1	0	0

Handwritten notes: Q_t (circled), Q_{t+1} (circled), J (circled), K (circled), Q_t (circled), Q_{t+1} (circled), J (circled), K (circled).

2. Perform Analysis of the circuit A given

State eq:

$$J_A = X \oplus Q_0$$

$$K_A = X \cdot Q_1$$

$$J_B = X + Q_1$$

$$K_B = X$$

$$Z = Q_1 \cdot Q_0 \cdot \bar{X}$$

State Table:

Q_1	Q_0	X	J_A	K_A	J_B	K_B	Q_1	Q_0	Z
0	0	0	1	0	0	0	1	0	0
0	0	1	0	0	0	0	0	1	0
0	1	0	1	0	1	0	1	1	0
0	1	1	1	0	1	0	1	0	0
1	0	0	1	0	1	0	1	1	0
1	0	1	0	1	1	0	1	1	1
1	1	0	1	0	1	0	1	1	1
1	1	1	1	1	1	1	0	0	0

State diagram

